	AIR UNIVERSITY
	DEPARTMENT OF ELECTRICAL AND COMPUTER ENGINEERING
	EXPERIMENT NO 3

Lab Title: **Verification Of Demorgan's Law**

Name: **Muhammad Waqas** **Reg. No:** **240756**

Objective: **To Verify the Morgan's Law .**

LAB ASSESSMENT:

Attributes	Excellent (5)	Good (4)	Average (3)	Satisfactory (2)	Unsatisfactory (1)
Ability to Conduct Experiment					
Ability to assimilate the results					
Effective use of lab equipment and follows the lab safety rules					

Total Marks: _____ Obtained Marks: _____

LAB REPORT ASSESSMENT:

Attributes	Excellent (5)	Good (4)	Average (3)	Satisfactory (2)	Unsatisfactory (1)
Data presentation					
Experimental results					
Conclusion					

Total Marks: _____ Obtained Marks: _____

Date: _____ Signature: _____

LABORATORY EXPERIMENT NO. 02

DeMorgan's Law

OBJECTIVE:

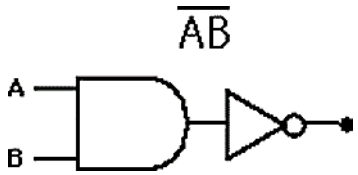
- To verify the DeMorgan's Law

Equipment required:

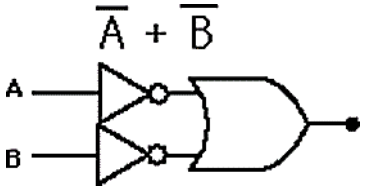
- TTL IC-7408
- TTL IC-7432
- TTL IC-7404
- TTL IC-7400
- TTL IC-7402
- Digital Electronics Trainer

First Law:

Left Hand Side:

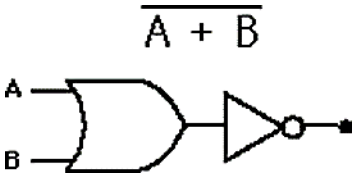


Right Hand Side:

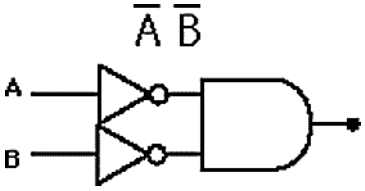


Second Law:

Left Hand Side:



Right Hand Side:



Lab Tasks:
Follow the given procedure and complete the given truth tables for both the laws.

Truth Table for DeMorgan's First Law:

Inputs		Truth Table Outputs for Each Term				
A	**B**	**A.B**	**(A.B)'**	**A'**	**B'**	**A' + B'**
0	0	0	1	1	1	1
0	1	0	1	1	0	1
1	0	0	1	0	1	1
1	1	1	0	0	0	0

Truth Table for DeMorgan's Second Law:

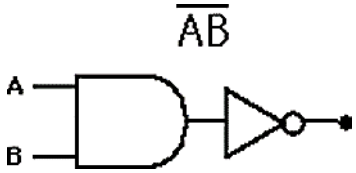
Inputs		Truth Table Outputs for Each Term				
A	**B**	**A+B**	**(A+B)'**	**A'**	**B'**	**A'.B'**
0	0	0	1	1	1	1
0	1	1	0	1	0	0
1	0	1	0	0	1	0
1	1	1	0	0	0	0

Simulation Activities:

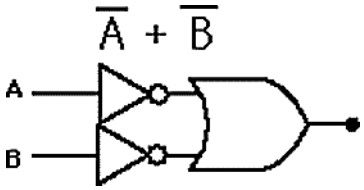
1. Simulate the given circuit for the verification of DeMorgan's first law.

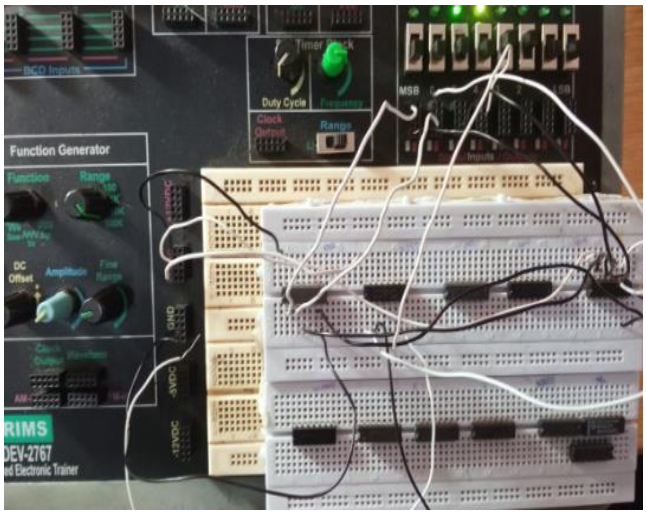
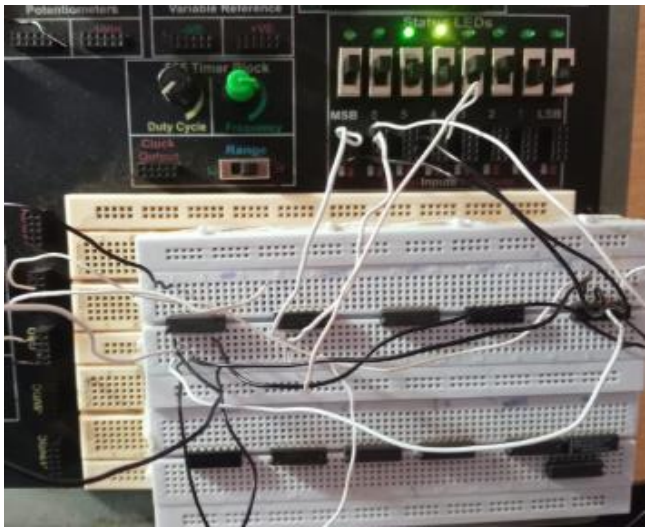
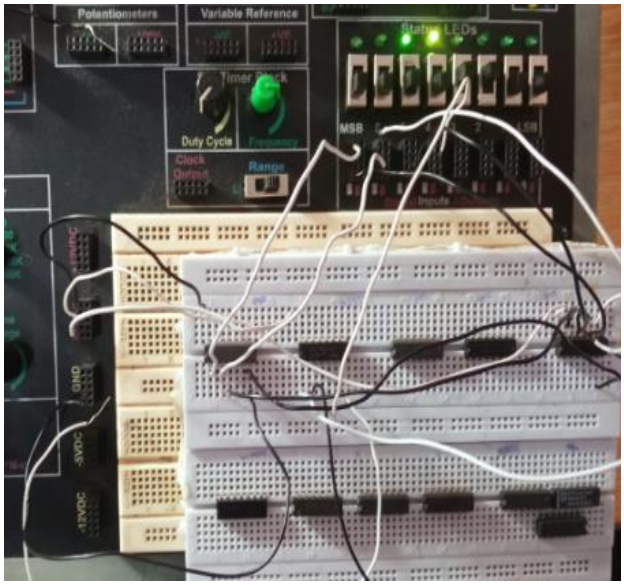
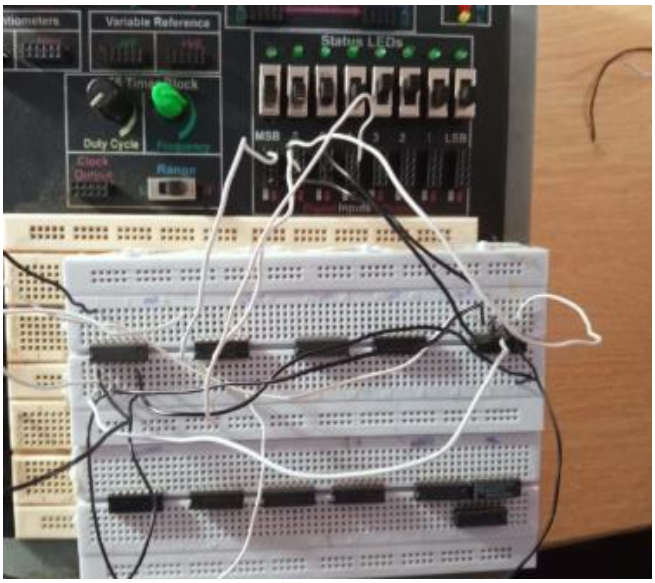
First Law:

Left Hand Side:



Right Hand Side:



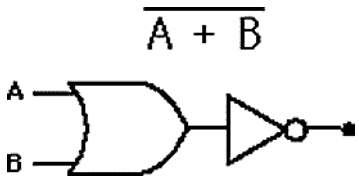
ON	**ON**
	
ON	**OFF**
	

Simulation Activities:

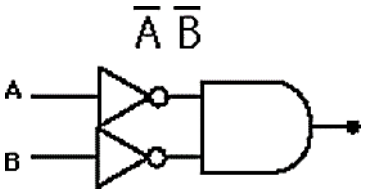
2. **Simulate the given circuit for the verification of DeMorgan's first law.**

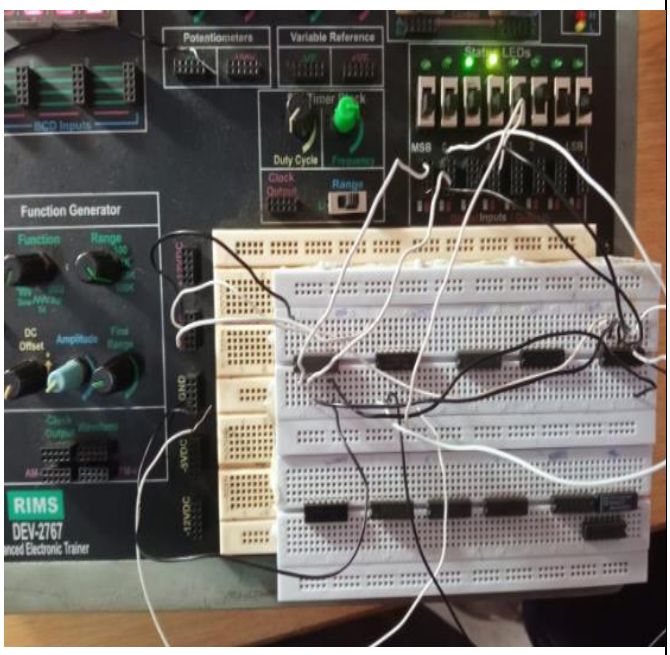
FirstLaw:

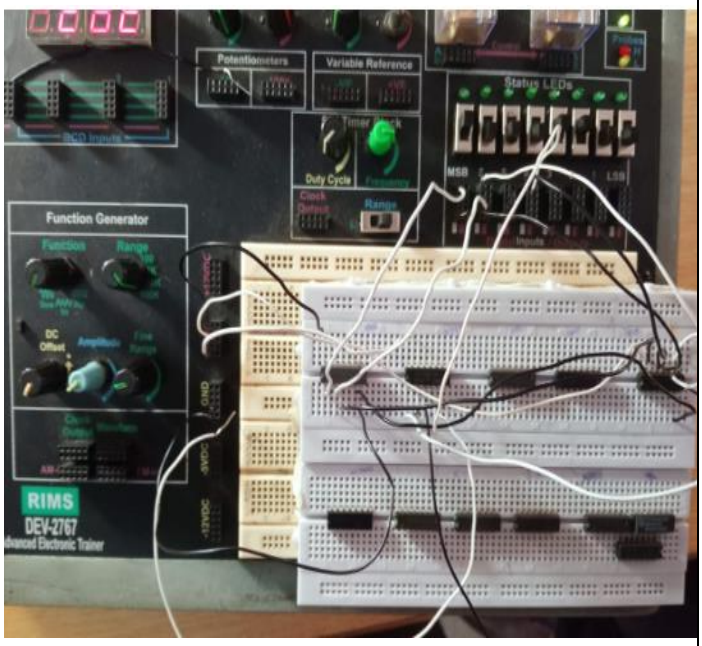
Left Hand Side:

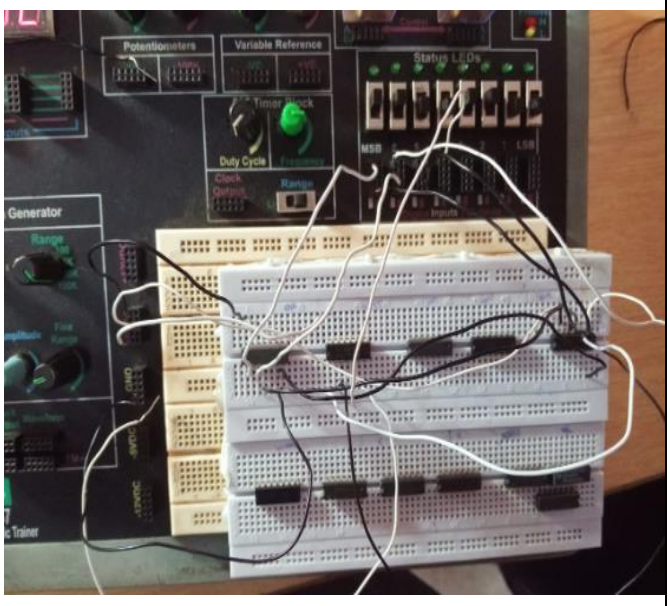


Right Hand Side:





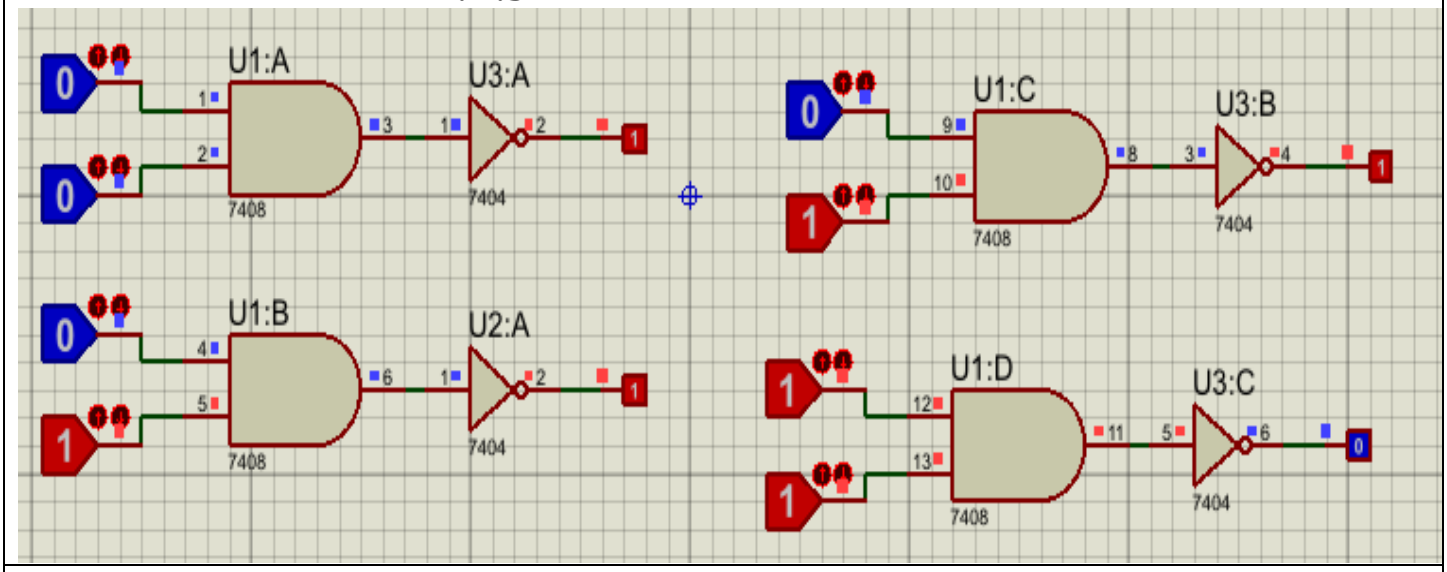




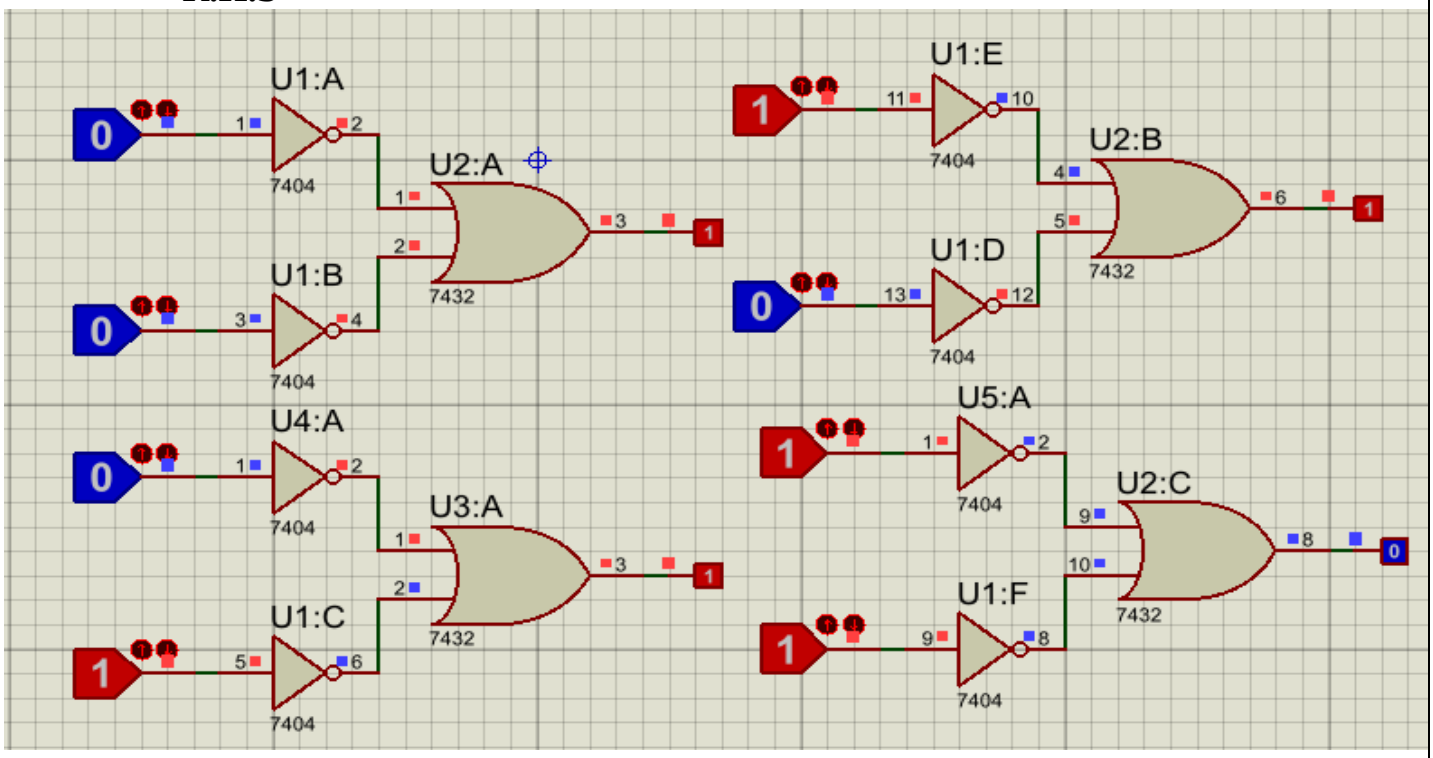


First law

L.H.S

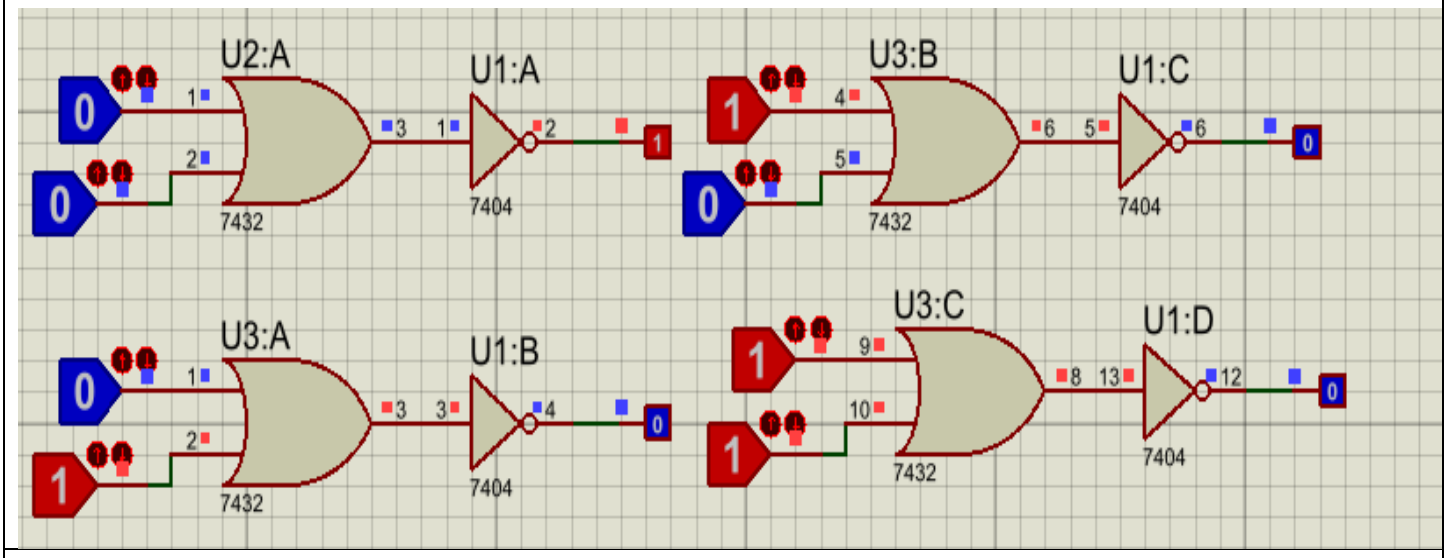


R.H.S

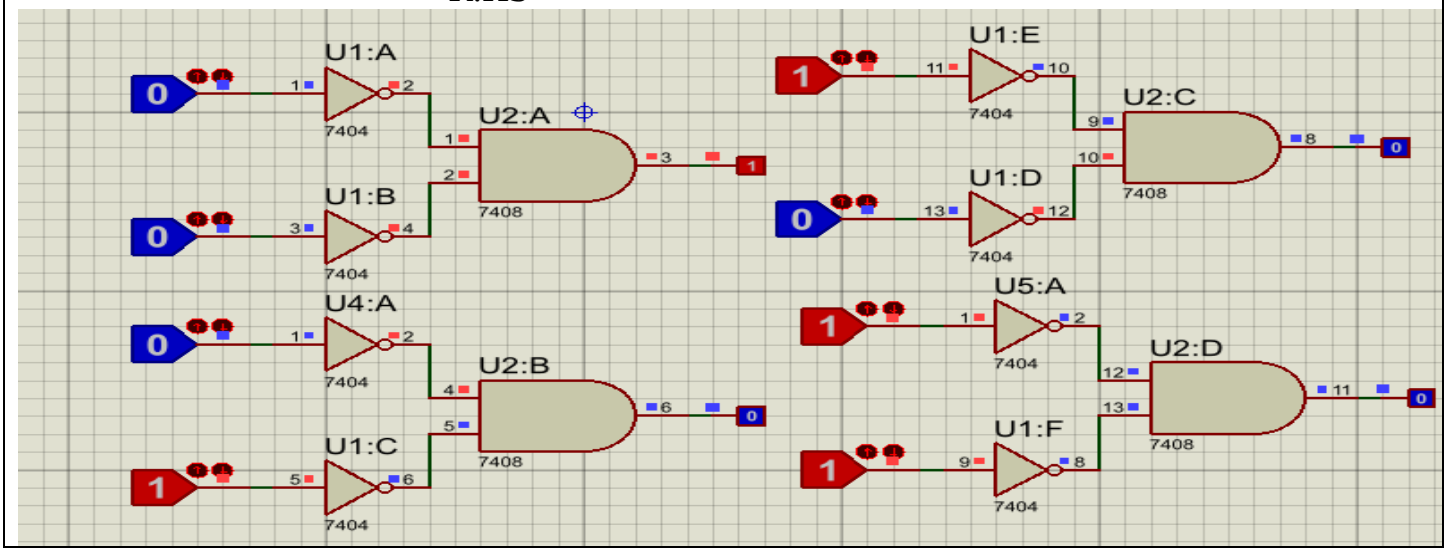


Second LAW

L.H.S



R.HS



Conclusion:

De Morgan's Law is very important in digital logic because it helps us simplify and understand how logic gates work. It states that the complement of a product is equal to the sum of the complements, and the complement of a sum is equal to the product of the complements. In simple words, a NAND gate can be used in place of an OR gate with inverters, and a NOR gate can be used in place of an AND gate with inverters. This shows that we can build the same logic circuits using different combinations of gates, which makes circuit design more flexible and cost-effective. De Morgan's Law is also very useful in Boolean algebra for reducing expressions and making circuits easier and faster. Overall, it provides a strong link between theory and practical use of logic gates in digital systems.